



Mechanistic insights into the initial explosion in the deflagration-to-detonation transition



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ABSTRACT

This work presents an imaging study of the complex and often overlooked early-onset mechanisms of the deflagration-to-detonation transition (DDT). Columns of granular octahydro-1,3,5,7-tetrazocine (HMX) doped with tungsten tracer particles are ignited thermally by hot wire and the resulting reactions are studied via high-speed X-ray radiography utilizing the Diamond Light Source synchrotron. The observed results provide insights into the initial development of the DDT process, resulting in a proposed mechanism for the slow initial steps in DDT for thermally ignited, low bulk-density granular explosive contained in a quasi-one dimensional tube configuration. In order to clarify the proposed mechanism, the terms *preferential flow channel* and *matrix burning* are adapted from the soil mechanics and hydrology literature to further elucidate the role of convective burning in DDT. The proposed mechanism helps to clarify the ongoing debate on the transport of gases in the Baer–Nunziato and reduced Baer–Nunziato models and suggests that applicability of each interpretation depends on which step in the mechanism is being described.

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1. Introduction

Accidents involving the detonation of large quantities of energetic materials can have grave real-world consequences, sometimes causing significant loss of life, destruction of infrastructure, and severe economic loss. Such accidents have occurred in both industrial and military settings, with the resulting explosions ranging from relatively small, for example the U.S.S. Forestal accident in 1967 [1], to the much larger Texas City disaster of 1947 [2]. The Texas City disaster remains the deadliest industrial accident in the history of the United States and resulted in the death of at least 581 people and the widespread destruction of Texas City. Although the hazards are well known, accidents involving explosives continue to this day; recent examples include the tragic events in Beirut in 2020, Port of Tianjin, China in 2015, and in the city of West, Texas in 2013. While many of the most prolific accidents involved ammonium nitrate and not military grade high-explosives, the gaps in our understanding as to the ultimate cause of DDT in both are similar.

A detonation is a supersonic shock wave supported by rapid exothermic decomposition of the explosive just behind the shock. Energetic materials designed to be detonable high explosives (as opposed to propellants or pyrotechnics) are generally initiated by the shock wave from a detonator. However, the most common accident scenario, which includes all the events mentioned above, starts with a simple thermal ignition that leads to deflagration, a much slower subsonic process that propagates via thermal and/or mass transport and, in general, is less violent than a detonation. Depending on a number of factors, a deflagration may develop into a detonation, and the overarching goal of this research was to gain a better understanding of the mechanism and tipping points that differentiate a smaller accident (deflagration) from a potential catastrophe (detonation). The process by which a detonation develops from a deflagration is referred to, self-descriptively, as the deflagration-to-detonation transition (DDT). As the primary mechanism leading to violent reaction following non-shock insults to explosives, DDT has been studied extensively and a thorough review of the subject by McAfee is provided [3]. However, due to the poorly understood, complex, multi-scale physical and chemical processes that drive DDT, it remains almost impossible to model either analytically or numerically and is still a very active field of research.

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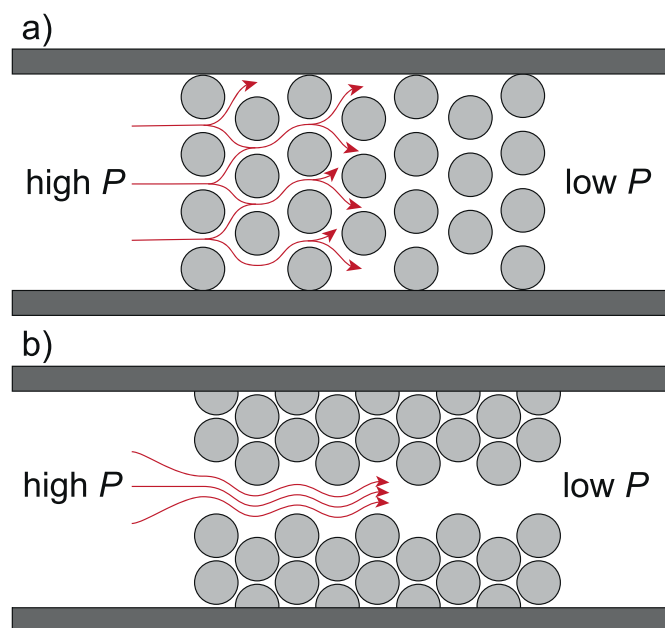


Fig. 1. Graphic highlighting the differences between a) matrix flow and b) preferential flow.

It can largely be agreed upon that for thermal ignition or cookoff, DDT is assumed to progress by the following simplified five-step mechanism:

- 1) The heat source ignites the explosive, causing surface burning.
- 2) In pressure-building systems the flame front penetrates through the porous network of the explosive via convective burning. Convective burning proceeds until the increasing pressure chokes flow within the pores.
- 3) The choked flow results in the formation of a high-density compact of explosive which is driven into undisturbed material by the pressure of the burning explosive.
- 4) The compression of the undisturbed porous material by the compact forms small, isolated, high-temperature regions (hot-spots) and ignites a compressive burn and a secondary explosion. This builds in pressure until a supported shock forms.
- 5) The shock builds in pressure until detonation occurs.

Within the DDT literature the phrase “convective burning” has been used to describe how hot and sometimes burning gases infiltrate an explosive bed. While the flow of hot gases through macro- and micro-pores can be termed “convective burning”, the distinction between these two appears to be important, at the very least, because of the differing amounts of surface area that gets incorporated into the combustion behind the convective burn front. As such we believe this phrase has been used imprecisely in the past to lump together the infiltration of burning gases through both macro- and micro-pores. To be clear on the exact meaning, this study uses established terms for flow in porous media from soil mechanics and hydrology [4]. The convention to describe flow in porous media in these fields are *matrix flow* and *preferential flow* [4]. *Matrix flow* refers to fluid flow through a bed of micro-pores resulting in an area-averaged infiltration front that appears to be spatially uniform, while *preferential flow* is used to describe flow through cracks or macro-pores and propagates more unevenly. The differences between these two flows can be seen in Fig. 1. For consistency, the infiltration of hot, reacting gases into micro-pores will

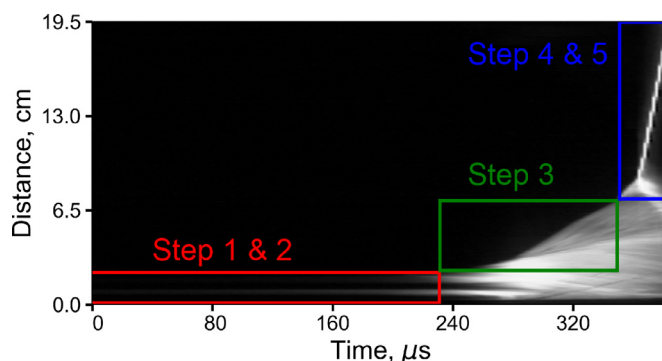


Fig. 2. Representative streak of a DDT experiment using 60% TMD HMX in an 8-inch polycarbonate tube. This experiment is shown in order to highlight where each step of the 5-step mechanism would occur. The red box highlights the slower processes (Steps 1 and 2 from the five-step mechanism) ($10^{-4} - 10^1 \text{ ms}^{-1}$). This area is the focus of this body of work. The green box shows (Step 3) the faster post initial explosion conditions ($10^1 - 10^2 \text{ ms}^{-1}$) and the blue box shows (Steps 4 and 5) the transition to detonation (10^3 ms^{-1}). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

be referred to as a *matrix burn* throughout. Given the proper definition, it is anticipated that many of the disagreements seen in the literature about the true nature of burn propagation contributing to DDT can be ameliorated [5–7].

Matrix flow transports fluid at a considerably slower rate than preferential flow due to a high degree of tortuosity in the porosity. As a result, even at the high pressures seen in the initial stages of DDT, the gases do not penetrate via matrix flow as rapidly or deeply as was once –and sometimes still is– thought [8–10]. Conversely, the typically wider, less tortuous crack systems seen in preferential flow can support fast moving flows of the working fluid deep into the bed before the micro-scale pore matrix can be significantly invaded [4]. Additionally, preferential flow can take on a variety of forms in granular materials, from a single channel to extensive interconnected branching networks [11,12].

Figure 2 highlights the steps involved in the DDT process and shows where each can be found in the 5-step mechanism above. The early time preliminary compaction and pressure build up (Step 1,2) have been far less studied than the later time high-density compact formation, acceleration, and eventual transition phase (Steps 3,4,5). In fact, the primary numerical models used today were developed beginning at step 3 using impact-ignition of the granular beds with initial velocities of $> 100 \text{ m s}^{-1}$ to simulate a high density compact traveling through the bed and in doing so ignore the slower processes that are present in a purely thermal ignition event. However, it was unknown whether the path to formation of a high-density compact would have an impact on the fate of the explosive. This has led to questions as to whether it was accurate to apply the current models to all DDT events even though a significant fraction of accidents begin with slower direct thermal insults and build up to the velocities seen in the impact-ignition experiments. Clarification into the physical phenomena of the early, slower processes of ignition, bed motion, and burn propagation and how it effects the later time stages for which the models were developed would help to clarify whether the assumptions made in the models are appropriate. The two primary models used today are the Baer–Nunziato (BN) model, which is a two-phase non-equilibrium continuum model, (where each of the phases contributes their own pressure, temperature, and velocity terms) and a reduced BN model, provided by Bdzil and coworkers, in which the velocity phases are locked [6,10]. These differ greatly in how they handle transport of gases throughout a granular bed: the BN model

relies on the convective burning infiltration in accordance with our description of matrix flow, whereas the reduced BN treats the gas infiltration into a stationary bed as an averaged bulk movement of the gas and solid phases. Furthermore, since both models are continuum models, they treat changes within the bed as a transverse average several lateral grains thick, creating a homogeneous zone, and therefore cannot be applied to many of the macro changes seen in the bed like the formation of preferential flow channels. As such, the focus of this work is solely on the initial slow processes (Step 1,2) that ultimately lead to the formation of the high-density compact (Step 3) and as we will demonstrate, the early time events appear to agree and disagree with elements from both of the primary models.

In this paper we will show that following an ignition event of a confined granular explosive with high porosity, surface burning leads to a building of pressure within the confining vessel. This results in a slight compaction (termed the preliminary compact throughout), and formation of a melt layer on the burning boundary of the explosive bed causing rapid restriction of matrix flow. The pressure eventually grows high enough to create a preferential flow channels leading into the uncompacted explosive, either by channeling through the melt layer and preliminary compact, or around the preliminary compact in a channel formed against the confining wall. The channel will then expose a region of uncompacted explosive with a much higher surface area and porosity, leading to matrix flow of the hot gases into the freshly exposed material. This causes a dramatic increase in reaction rate, that we call the initial explosion, as the hot gaseous products start a matrix burn in the region. Evidence for the initial explosion can be found throughout the literature dating back to the pioneering works by Bernecker and coworkers [9,13–15]. However, as ubiquitous as an initial explosion is to the build up in DDT, the importance of it has yet to be fully explored or described in the literature due to historic difficulties in capturing the process in sufficient detail.

A considerable number of valuable insights into the dynamic processes of energetic materials have been made using X-rays and proton imaging in recent years, significantly advancing the DDT field [16–22]. Sparse radiographic image sequences and optical streak records such as these have been relied upon to formulate the current mechanisms used for DDT in tube confinement. In general, tabletop X-ray sources have lacked the ability to simultaneously resolve both temporally and spatially some of these dynamic processes. However, synchrotron light sources are now routinely available for high-speed imaging experiments. The achievable spatial resolution, temporal resolution, field of view, and recording time depends on the facility. For our work, we used the I12 Beamline at Diamond Light Source, UK [23]. Radiographs were taken at a frame rate of up to 37 kHz, with a nominal resolution of 17.9 microns/pixel. These radiographs provided multi-dimensional details of reacting energetic materials that have never been observed before. While an improvement over tabletop X-ray sources, the time-scales observed are still inadequate to allow for tracking of the reaction from the ignition event all the way up to DDT.

The work presented is focused only on the slower phenomenon (Steps 1,2 of the simplified five-step mechanism above) that build up to and cause the initial explosion to occur. We attempt to demonstrate that the establishment of an initial explosion (and the power of the ignition source) can affect the ultimate outcome. Through defining the early time steps that are often overlooked, we anticipate that we can combine our results with recent research that focused on the faster, later stages of DDT, thereby providing a more complete understanding of the entire 5-step process. For example, Tringe and coworkers recently described novel phenomena for the faster rate, impulsively-ignited regimes of granular explosives [22]. This work revealed the first evidence of preferential flow channels forming and leading to an asymmetric bypass of

a compacted region of explosive in the fast stages just before DDT occurs. Since this work utilized an impulsive, thermite-driven ignition system which mimics the initial explosion, it could only allow for steps 3–5 of the simplified five-step mechanism above to be interrogated. Similar to thermite ignition, historical DDT tests using impact pistons also begin at step 3 since the plug formed by an initial explosion is created by the piston. Importantly, the BN model was built off of the impact ignition data and as such does not account for the first two steps. It has been seen in previous research [16] from our lab and unpublished data from Tringe and coworkers that ignitions that start at step 1 (slower thermal) don't always lead to detonation transition. However, if the progression of step 1 and 2 creates a powerful initial explosion, then the HE becomes compacted and accelerated in a manner consistent with piston impact; that is, it accesses the more well defined step 3.

With the ability to record hundreds of thousands of images at up to 37 kHz at the Diamond Light Source, we were able to interrogate this gentle ignition process that occurred over long time scales. This would have been difficult to accurately time and record using high-power flash X-ray heads like those which were used by Tringe and coworkers or light sources which deliver photons in discreet packets. Further diagnostics included tungsten particle tracers to perform particle tracking and digital image correlation (DIC). The radiography, particle tracking, and DIC were used to illustrate the initial surface burn, preliminary compaction, and formation of the initial explosion. The data obtained was then compared to visible high-speed videography taken at Los Alamos National Laboratory (LANL) using identical test setups. In all cases, it was found that an initial explosion originates within the distal extent of a preferential flow channels formed in the bed of explosive. Furthermore, we have found that the insights gained by directly examining the radiography and visible imaging in this work allows for new information to be extracted from legacy visible-only streak images.

2. Experimental

A large body of work exists which demonstrates how ignition sources, porosity, and other variables may lead to varying mechanisms for the initial stages of DDT [24]. As such, the work described herein focuses only on gasless, slow thermal ignition of low- to moderate bulk-density granular explosive beds of HMX (cyclotetramethylenetetranitramine) that contain high porosity as the proof-of-principle model system. Furthermore, these test were performed using weak confinement, which is defined as confinement that will deform under the pressures produced during ignition. HMX was chosen as the model system since it can be used in small quantities, making it more amenable to the size constraints of the diagnostic methods developed in this work. Ignition of the explosive beds were performed via electrically heated NiChrome hot-wire as it is an appropriate surrogate for a slow direct thermal insult like that seen in an accident involving a fire with surface ignition of a predominantly cool explosive material mass. CHEETAH calculations show that the products produced through reaction of HMX produces 10 mol of gas per 1 mol of solid leading to a rapid pressurization of the fixtures [25].

Polycarbonate and aluminum tubes with an outer diameter of 25 mm were cut to 30 mm lengths and a 3 mm hole was bored down the center (Fig. 3). Four 600 μ m holes were drilled perpendicular down the length of the tube to allow fiber optics to be inserted. These fibers were used to gather information about the arrival time of visible light. The tubes were capped by steel endplates and bolted together to ensure pressure would build within the vessel. The reaction was initiated by applying current to a small loop of NiChrome wire inserted into the end of the tube.

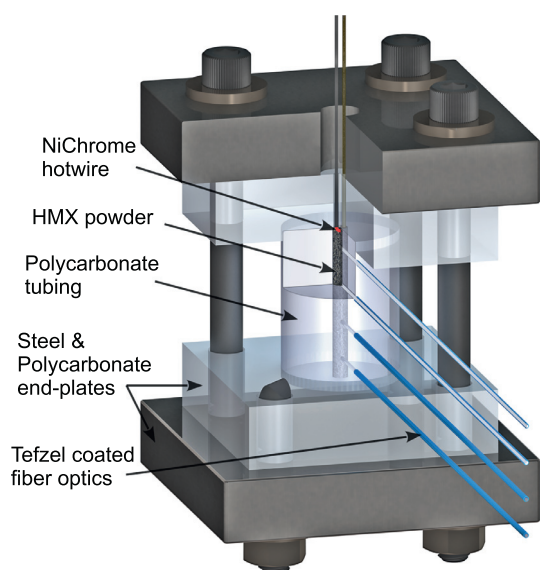


Fig. 3. Detailed rendering of the reaction vessel.

For all experiments, Type 2 HMX powder (Class 2, Grade B: MIL SPEC MIL-DTL-45444 Rev. C) purchased from Chemring Nobel was doped with 2% by weight, 100 μm tungsten particles to allow for monitoring the movement of the explosive bed. Type 2 HMX at STP has a burn rate of 0.3 mm s^{-1} , and a melting point 550 K and tungsten has a melting point of 3695 K. The difference in densities between the tungsten particles and explosive allowed for discrimination of the tungsten particles by radiographic imaging. The size of the tungsten particles were selected to be of a similar size to the HMX particles. This allowed for the particles to remain locked with the HMX in the low acceleration regimes studied in this work. Additionally, with an imaging resolution of 17.9 microns/pixel, the 100 μm tungsten particles allowed for sufficient resolution of the particles. Three bulk densities of granular HMX explosive were analyzed for this research; 60%, 70%, and 80% of theoretical maximum density (TMD)(TMD for HMX: 1.91 g cm^{-3}). It is assumed for the analysis that the tungsten particles were locked in with their neighboring HMX particles resulting in all particles flowing in a similar manner. While the high explosives were visible without the tungsten tracer particles, the inclusion allowed for enhanced data analysis techniques to be used, such as particle tracking and DIC. The Python based particle tracker TrackPy version 0.4.2 was used to track the tungsten particles and provide the data needed for the velocity field calculations [26]. Since the flow could be approximated as one-dimensional the ChannelPredict functionality was used with the size of the tungsten particles being in the 15 pixel range. The muDIC Python package, version 1.0 was used to implement Digital Image Correlation (DIC) in order to calculate the evolving strain field of the HMX bed.

Beamline I12-JEEP (Joint Engineering, Environmental, and Processing) at the Diamond Light Source Ltd (UK) was used. A 73 keV beam was used to provide the maximum contrast and resolution of the tungsten particles. The images were recorded at 10 kfps and 1280×256 pixels using a Vision Research Miro 310M high-speed camera coupled to a 700 μm thick, single crystal of LuAG:Ce (Lutetium Aluminium Garnet doped with Cerium) X-ray scintillator. The nominal imaging resolution was 17.9 microns/pixel. Identical experiments were performed at LANL but were interrogated using high-speed videography (See Fig. 2 for example). A Vision Research

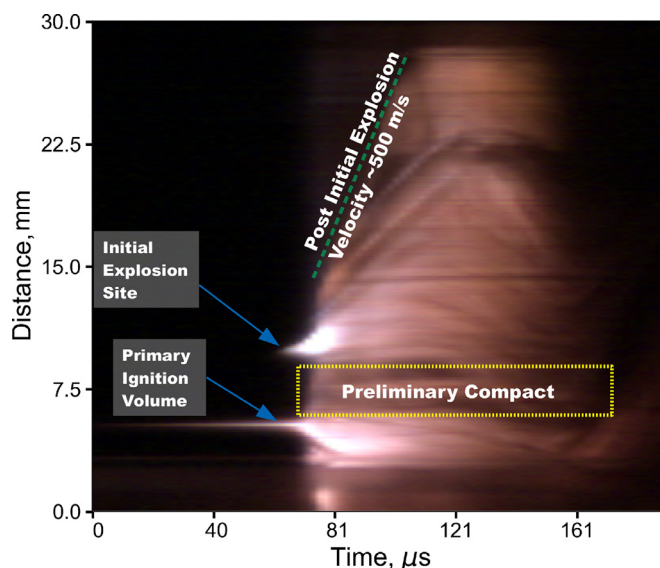


Fig. 4. Distance versus time streak captured by high-speed optical imaging. Arrows and boxes highlight the distinct regions where the primary ignition site, preliminary compact, and initial explosion occurs. This streak was made using an identical tube geometry and setup as was used for all the work performed at the Diamond Light Source. The dark zones typically seen are indicative of the pre-ignition volume at the beginning and at later times of the preliminary compact (yellow box). The focus of this work is on the often overlooked 3–5 m s^{-1} particle velocity observed during the development of the preliminary compact and the initial explosion site. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Phantom 2512 high-speed video camera was used to image the reaction.

3. Results and discussion

Results obtained using high-speed videography provided a visible-light 1-D DDT streak record for direct comparison to the radiographic images recorded in this work (Fig. 4). Historically, 1-D visible-light streak records have been important in the design and development of the predominant mechanisms and models still used today for DDT. Unfortunately, as we will demonstrate, the visible-light videography of the burn surface masks many of the important phenomenon occurring within the energetic material bed. This is due to the streak records obtained only containing a one-dimensional interrogation of a surface burn (specifically only the burning occurring at the explosive confinement interface) while there are important 3D processes at work. For instance, due to the heterogeneity of the reaction within the explosive bed, the streak information can often tell conflicting stories by simply choosing a different azimuth to streak. These limitations have led to many of the discrepancies seen within the common simplified 5-step mechanism described in the introduction.

In this work it was found that a more accurate and complete picture could be obtained by combining the spatially one-dimensional streaks with two-dimensional DIC, and radiographic imaging obtained from the Diamond Light Source. The mechanism proposed at the end of this manuscript was developed through the use of this suite of diagnostics and analysis techniques. Only the 60% TMD replicates are shown since similar behaviors were observed in all other densities tested. Two of the 60% TMD HMX samples tested indicated a preliminary compact forming in the bed around 5 ms prior to an initial explosion. The remaining two samples did not show a defined timepoint where the preliminary compact occurred, but instead showed a more homogeneous compaction over a longer period prior to an initial explosion.

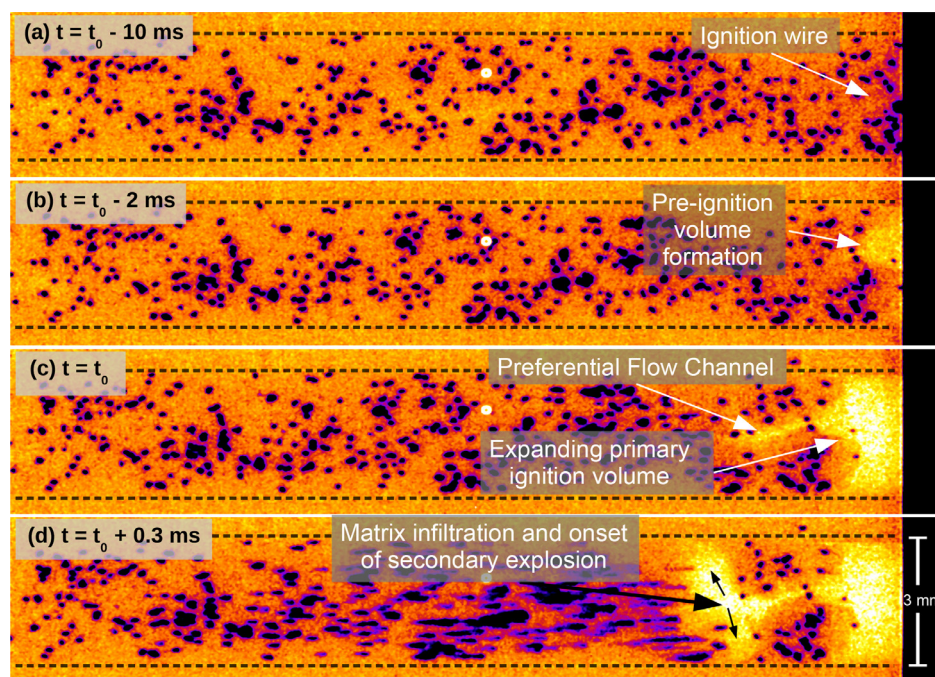


Fig. 5. Radiographic image sequence of a 60% TMD HMX test showing the formation of a large preferential flow channel down the center of the preliminary compact of the HMX. Frame 1 was taken prior to powering the hot-wire. Frame 2 shows the formation of a pre-ignition volume around the hot-wire. Frame 3 shows the primary ignition volume and the opening of a preferential flow channel through the preliminary compact. In the final frame an initial explosion is evident. False colors allowed for easier visualization of the materials. The bright spot that appears in the center of all the images was a defect on the X-ray scintillator. Dark spots are tungsten particles for tracking movement of the energetic material. Color range from yellow to purple represent areas of low to high mass respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

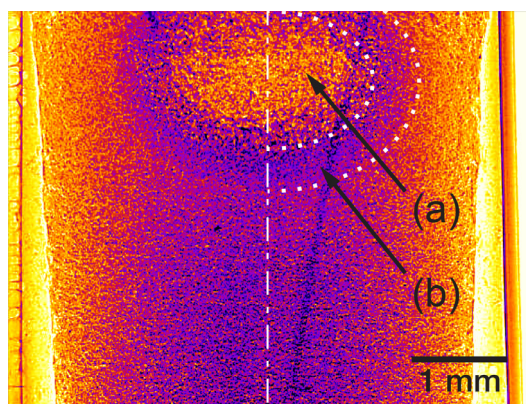


Fig. 6. Radiographic image of a small 4 mm diameter cylinder of energetic material heated to thermal runaway and auto-ignition. (a) shows clear evidence of the formation of a pre-ignition volume while (b) highlights the melt layer surrounding it. Color range from yellow to purple represent areas of low to high mass cross-section respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.1. Proposed mechanism

Here we will propose our detailed mechanism expanding on steps 1 and 2 of the simplified mechanism shown in the introduction and then spend the remainder of this discussion describing the evidence for each step in detail. It is proposed that the mechanism for development of the initial explosion that ultimately leads to DDT consists of the following steps.

- 1) The source, either from external stimuli or from self-heating, begins melting and decomposing the explosive resulting in the production of gas and formation of a pre-ignition volume. The gas-phase reaction produces additional heating, melting and

decomposition of the explosive. The now-molten explosive seals the porosity which would have allowed the evolved reactive gases to permeate into the explosive bed via matrix flow.

- 2) The gas pressure continues to build, compressing the explosive, enlarging the pre-ignition volume, and beginning the formation of a preliminary compact. As the pre-ignition volume grows the melt layer formed around it expands. The melt layer in combination with the preliminary compaction of the explosive continues to suppress the matrix flow of the gases, leading to a rapid increase in pressure and temperature, and ultimately the development of a primary ignition site. The development from a pre-ignition volume to the primary ignition site is defined by the point at which thermal runaway (self-sustaining reaction) occurs. This is commonly referred to as combustion and luminous flame becomes apparent at this stage. Importantly, if the external heat source is removed at this point the reaction would continue and proceed to the next steps.
- 3) The pressure eventually grows high enough that a preferential flow channel is created through the melt layer and preliminary compact or between the confinement walls and the preliminary compact.
- 4) Either of these scenarios leads to the formation of a preferential flow channel that exposes fresh explosive downstream, containing a greater surface area than that seen in the primary ignition volume, to reactive gases at high temperatures and pressure. This leads to a resumption of matrix burning, resulting in a drastic increase in reaction rate, which is termed the initial explosion. From here we access step 3 of the simplified mechanism in the introduction.

3.2. Formation of the pre-ignition volume and melt layer

Figure 5 clearly demonstrates one of the common paths that leads to an initial explosion. The first frame shows the vessel

just before the NiChrome wire was heated. In the second frame a small pre-ignition volume can be observed forming and expanding around the hot-wire. When visualized with an optical diagnostic as verified in Fig. 4, the dark pre-ignition volume develops into the primary ignition volume at which point it would be seen filled with hot, luminous gases.

The melt layer surrounding the pre-ignition and primary ignition volumes is not directly observed in this experimental series, due to the lower resolution of the X-ray. However, based on previous work in the literature and high-resolution experiments performed by our group at the Diamond Light Source beamline I13-2, as seen in Fig. 6, a melt layer forms around these ignition volumes [18,27]. Even with differences in ignition methods, whether ignited using a hot-wire or by thermal runaway, it became apparent that the pre-ignition volume forms a melt layer of explosive on the boundary of the volume. Further evidence of the presence of this melt layer was verified by quenching the reaction via a release of pressure at early stages. When this was performed, we observed a resolidified surface melt layer forming a hardened concave shell around the quenched primary ignition volume. This observation has also been demonstrated in HMX strand burner experiments by Taylor and coworkers, where the explosive appears to be uncovered slowly enough that the heat from the reacting gases maintains the integrity of the melt layer [28]. This melt layer is anticipated to play a vital role in the suppression of matrix flow of reactive gases into the explosive bed.

3.3. Formation of the preliminary compact

In addition to the melt layer, formation of a compacted region can suppress matrix burning as well. At this point the full BN model would become redundant and the reduced BN formalism would be more computationally efficient. As the gas pressure builds in the vessel, the enlarging primary ignition volume is opened across the entire diameter of the tube as seen in frame three of Fig. 5. While the explosive is still sealed by the melt layer, the HMX continues to thermally decompose producing reactive gas-phase products. These reactive products flow into the primary ignition volume where they further react in a highly exothermic fashion. The heat from the exothermic reaction feeds back to the melt layer, further melting and decomposing the HMX producing more reactive products. In the steady-state this is similar to combustion of nitroamines as described in Propellants and Explosives [29], which is commonly referred to as a laminar burn in the DDT community. Particle tracking analysis of the entrained tungsten particles show that the presence of increasing time-variant pressure compressing the granular bed leads to a gradient in the velocity of the particles, where the particles closer to the burning surface of explosive are moving faster than the ones further downstream. Intuitively, this leads to the formation of a density gradient in the explosive bed, where the material closer to the burning surface undergoes more compaction than the material further down the tube. Indeed, it can clearly be seen that both a particle velocity and a density gradient are established from our analysis. As observed by Fifer and coworkers, the burning of an explosive in a supported tube leads to the formation of a preliminary compact and consequent suppression of matrix burning until significantly higher pressures are reached [30].

This data set was also amenable to DIC due to the inclusion of the tungsten particles. DIC allows for the tracking of strain, which we interpret to be compaction of the bed from permanent plastic deformation. When coupled with the velocity fields of the beds DIC could be used to further strengthen the proposed mechanisms by demonstrating that the majority of the compaction occurs at the end of the explosive column adjacent to where the burn is occurring. Figure 7 shows the evolution of what was seen in a rep-

resentative example of the DIC analysis performed for one of the 60% HMX samples. The strain maps created were found to be consistent across all the experiments. They suggest that rather than the bed of HMX crystals compressing in a uniform manner, which is what would be deduced from the one-dimensional streak images, the bed collapses in a localized fashion forming the preliminary compact. Further, it appears that once the localized bed collapse starts to occur, the strain increases in the same area. While the exact location of the bed collapse appears to be random, the occurrence of the phenomenon is consistent.

3.4. Formation of preferential flow channels and initial explosion

There are several factors which influence the rate and magnitude at which the pressure can build within these experiments. One of these factors is the formation of cracks within the explosive bed and the effect that this has on the reaction rate. Once the pressure exceeds the strength of the weakest point within the explosive bed, a preferential flow channel can be opened (as seen in the third frame of Fig. 5). Frame three clearly shows evidence of a preferential flow channel formed through the preliminary compaction. This channel opens a less restrictive path for the hot, high-pressure gases to access and react with the less compacted explosive which is not yet shielded by a melt layer. When flames penetrate into cracks of uncompacted material a subsequent step function increase in pressure and burn rate is often seen. This amplification is the initial explosion and can be seen in frame four of Fig. 5. The occurrence of the initial explosion is indicated by the small channel that has opened into the preliminary compact, the relatively stationary particles to the right of the breakout, and the rapid leftward acceleration of everything further downstream. With visible diagnostics this stands out as a region of increased light intensity.

While frame three of Fig. 5 appears to be clear about the development of preferential flow channels, differences discovered between replicates establish that the formation of preferential flow channels is quite complex and can take on other morphologies. For example, as was shown in Fig. 5, the formation of a preferential flow channel is seen developing down the center of the explosive bed. At the apex of the primary ignition volume a preferential flow channel develops allowing gases to bypass the densified region at the head of the preliminary compact [31].

An alternative morphology often seen in this work is displayed in Fig. 8. As can be seen, this experiment led to the formation of a preferential flow channel along the inner wall of the tube. The particles at the edge of the preliminary compact are seen being ejected rearward into the burning material in the primary ignition volume. The injection of material increases the pressure until hot gases can intrude into the channel along the wall of the tube. As with the morphology of a preferential flow channel through the bulk of the preliminary compact, a channel formed along the confining wall also acts as a preferential flow path and leads to the commencement of the initial explosion once the pressure is forceful enough to bypass the compact. Regardless of the preferential flow path morphology, the final result is ultimately the same. The only difference seen is in the time required for the hot gas to be able to bypass the preliminary compact and melt layer accessing the less compacted bed conditions needed for a high-mass burn rate initial explosion.

Currently the majority of research on DDT has been performed in a one-dimensional configuration. While these one-dimensional tests have provided a wealth of information and have directly led to many of our current theories on DDT, such as the simplified 5-step mechanism outlined previously, most DDT configurations of interest will not be strictly one-dimensional in nature. Effects such as the dispersive nature of shocks in higher dimensions,

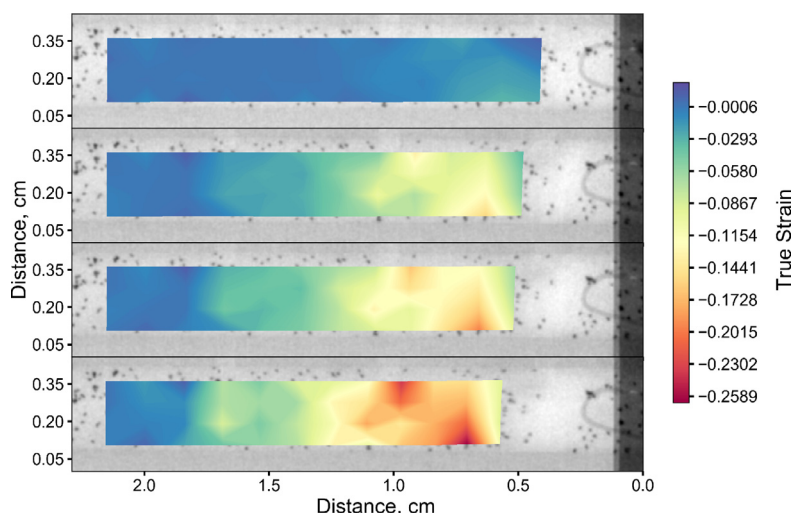


Fig. 7. DIC results showing the time evolution of compaction of a typical HMX column seen in this work. The color map is proportional to the true strain on the HMX bed. Blue represents ambient and spans to red which represents the local maximum within the system. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

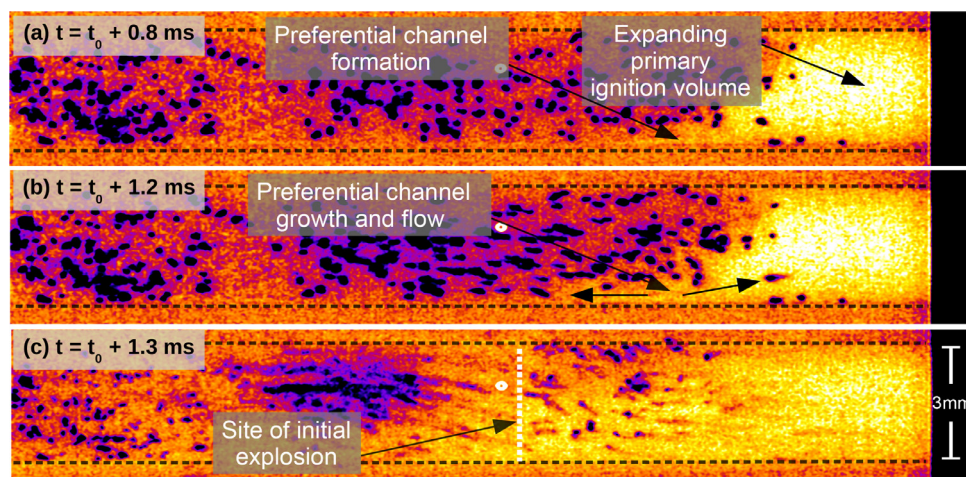


Fig. 8. Evolution of a preferential flow channel down the side of the tube. Dark spots are tungsten particles for tracking movement of the energetic material. Color range from yellow to purple represent areas of low to high mass cross-section respectively. The low density yellow region is likely filled with luminous flame. a) Shows the formation of the channel down the bottom edge of the tube. b) The channel continues to grow. Blurring of the tungsten particles indicates motion of the particles, where the motion would be in the direction that the narrow side of the blurred particle points. The backwards movement of the particle at the mouth of the channel, as well as the forward movement of the particles at the tip of the channel suggests that the pressure is increasing in the preferential flow channel. c) As indicated by the motion blurring of the tungsten particles, the initial explosion occurs approximately halfway down the image (dotted white line), bypassing the compacted HMX. Dashed black lines show the location of the tube walls. The bright spot that appears in the center of all the images was a defect on the X-ray scintillator. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

wave interactions, and potentially undiscovered phenomena suggest that the simplified 5-step mechanism may not be the only way to achieve DDT. For example, the weak confinement tests performed by Tringe and coworkers suggests that a new, currently unexplained, mechanism is responsible for the transition to detonation as the experiment deviates from a strict one-dimensional configuration. However, the mechanism explored in this work is inherently three-dimensional and it is expected that the formation of the pre-ignition and ignition volumes, preferential flow channels, and initial explosion will occur in a similar fashion in two- and three-dimensions.

As with the morphology of a preferential flow channel through the bulk of the preliminary compact, a channel formed along the confining wall also acts as a preferential flow path and leads to the commencement of the initial explosion once the pressure is forceful enough to bypass the compact. Regardless of the preferential flow path morphology, the final result is ultimately the same.

Tying all of these findings together, it becomes apparent for the first time, that the initial slow processes ($\text{mm} - \text{cm s}^{-1}$) studied here are similar to the fast ($10\text{--}100 \text{ m s}^{-1}$) late stages of compact formation and asymmetric bypass in weak confinement seen recently by Tringe and coworkers [22]. In related fashion in their work as well as ours, a locally dense compact is formed in both scenarios and is driven down stream by increasing pressure. Once the pressure builds enough to force the formation of a preferential flow channel, hot gasses can infiltrate ahead of the compacted region leading to a matrix burn and a subsequent acceleration in the reaction rate. In the slow, early phase case as studied in this work, the matrix burn penetrates only a short distance (on the scale of a charge diameter) and is quickly suppressed due to secondary compaction of the explosive by the initial explosion [9]. After pressure builds and accelerates this second higher-density compact (commonly referred to as “the plug” in many archival publications on DDT mechanisms), it can then undergo asymmetric bypass (via

preferential flow channels through or around the compact), where the hot gases infiltrate the porous material in front of the compact and, if there is enough energy released, undergo the transition to a detonation. From the combined efforts of this work and the work by Tringe and coworkers it becomes clear that the lead up to DDT may be a multi-plug process. Our work shows that a plug must be formed and bypassed prior to beginning step 3 of the simplified 5-step mechanism. The bypass of this initial compact, while it does lead to an intensification and the initial explosion, it is insufficient for DDT to occur. However, it appears that following this, in steps 3–5, a new faster moving plug is formed that can lead to a DDT as described by Tringe. However if the energy is insufficient, another compact forms and the process repeats until transition is achieved or the reaction is exhausted. The first reported evidence of a multi-plug process was shown by McAfee, but it was not clear at the time why this was occurring or the significance of it [7]. Future research will attempt to implement an upgraded set of diagnostics to further bridge the gap between these slow and fast regimes. This will be used to develop precise models to describe each of these distinct pathways more accurately in an effort to understand how they converge on the same outcome.

4. Conclusions

In this work, we developed a novel workflow and diagnostic suite utilizing the Diamond Light Source. This allowed for multiple important insights into the early stages of the DDT process in porous granular beds of HMX to be discovered. Furthermore, the results appear to clarify an ongoing, multi-decade disagreement about the role of convective burning in the BN model and the coupled reduced BN proposed by Bdzil that is not burdened with multi-phase velocities. This work shows that neither positions were exactly right, but both contained elements that were correct. It can be seen that the BN model is most accurate before and during the initial explosion. This is intuitive since the high density compact that is formed as a result of the initial explosion acts to cease all matrix flow. In the stage just after the initial explosion, where the preliminary compact and melt layer have ceased matrix flow but before the preferential flow channel has opened, the reduced BN model is more accurate. This is valid as under low gas flow velocity regimes, the melt layer makes up the difference to keep a large drag assumption valid [32]. However, as both BN and reduced BN are continuum models, neither can fully capture the physics of and effects due to the non-1 dimensional formation of a preferential flow channel propagating combustion through or around compacted sections of explosive. It appears as though a hybrid model is needed as it would allow for a more accurate description of the early stages of DDT.

Some of the most interesting findings came from a combination of the particle tracking and DIC. These strongly suggest that, in the pre-ignition volume and at the primary ignition site, matrix flow within the bed is halted by the formation of a thin melt layer and the preliminary compact. This allows for the pressure to grow behind the preliminary compact until a preferential flow channel is formed or extended into new higher surface area material via either a weakness in the center of the column or between the wall of the confinement vessel and the explosive bed being exploited. This then allows for the energetic material to undergo the initial explosion and kicks off the more well studied downstream high-density compact acceleration that finally leads to DDT.

As a result of the new insights gleaned from this work, it is proposed that the mechanism for the development of the initial explosion that leads to an environment conducive to DDT be clarified and defined as laid out in Section 3.1. Ultimately, this will help to develop a framework for the tools needed to solidify the DDT mechanisms for thermally ignited granular beds of explosive ma-

terials of interest in weak confinement. In future work, other energetic materials (ammonium nitrate, porous polymer-bonded explosives (PBXs), etc.) and alternate ignition scenarios will be explored to determine whether similar phenomenology exist. Refined descriptions of the physics incorporated in both the slow and fast regimes of the DDT process has the potential to produce insights that could allow for identification of risk and mitigation of some of the most common types of disasters caused by energetic materials.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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